

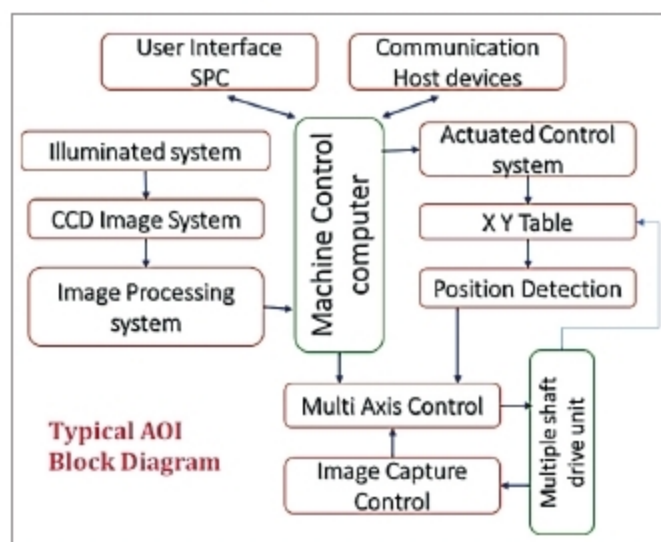


Fig. 2: Typical AOI block diagram

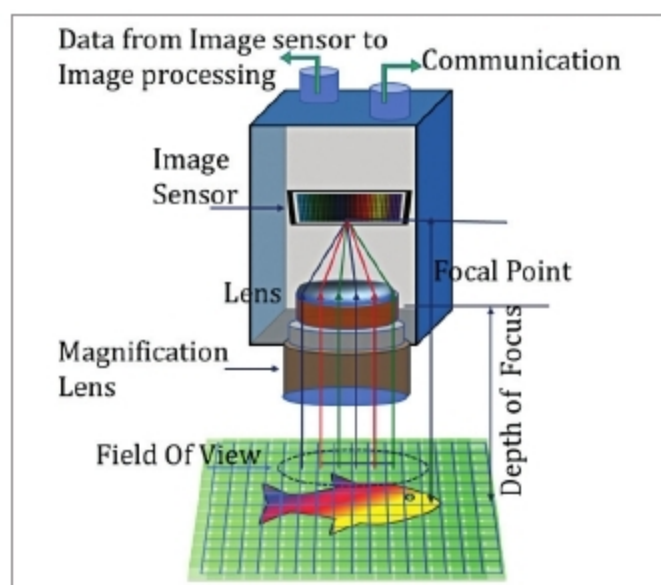


Fig. 3: CCD image system

Camera Resolution	15μm	20μm
FOV size	30mm X 30mm	40mm X 40mm
Inspection Speed	22.9 – 36.6 cm ² / sec	
Camera Speed	4 mega pixel high speed camera	
Illumination	IR – RGB LED dome styled	
Max measurement height	5mm	

Fig. 4: Specifications

- Solder fillet
- Missing lead
- Bridging
- Markings

Fundamentals of AOI

The core structure of the automatic optical inspection system includes a CCD image system, AC servo control x, y workbench, and image processing system. In other words, the CCD image system, AC servo control x, y workbench, and image processing systems are the core of AOI. The actuating parts (drivers and axis), image grabbing system (sensors, optical and illumination), and software processing part work in close relationship to each other, so the speed of each must be synchronised.

With the development of digital computing, machine vision and image processing have become essential to automate the inspection process with the help of high-resolution cameras, lighting devices, illumination techniques, and an efficient processing algorithm. The faster the processor, the less time the machine vision system will need to process image information. Speed is an important factor for high-volume production lines. Also, specific inspection software is available as per specific needs.

The block diagram of the AOI system is given in Fig. 2. This is the same for solder paste inspection. The only difference is light source or illumination and processing software. These machines differ in terms of the height they need to measure. SPI measures the height of solder paste, whereas AOI measures the height of components. Therefore AOI has a four-way or eight-way projection, while SPI is provided with a two-way or four-way projection.

Illuminated system

For any type of camera, enough light means a guaranteed good image. Lighting is important because it illuminates the part to be inspected. Many types of lighting sources such as fluorescent tubes, UV tubes, strobe lamps, high-intensity halogen lamps, and LED lamps are being used in the electronics industry for the best illumination.

Currently, many AOI manufacturers are using LED lamps and IR-RGB LED dome styled illumination. The choice of a light source depends on what you want to inspect. The lighting system generally includes white, red, green, and blue lights in a configurable lighting module. Machine vision systems need to carefully control lighting or adjust light intensity to ensure accurate and repeatable results by using digital controllers. The angle of lighting is also important because taller components in PCBA block the light from reaching smaller components.

Hence, the latest AOI machines are equipped with one or more projections

from different angles to offer the best overall illumination. Also, the selection of the type of light and the location at which the light source is fixed matters a lot.

CCD image system

Typically, five components make the machine vision system (Fig. 3).

1. Lighting
2. Lens
3. Image Sensor
4. Vision processing or image processing
5. Communication

Lighting illuminates the part to be inspected. The lens captures the image and presents it to the sensor in the form of light. The image sensor converts light into a digital image, which is then sent to processing unit for analysis.

Image processing follows a process to review the image and extracts the required information as per the interest of the user. Finally, the resulting data is communicated as per the user's interest.

Before proceeding further into the vision camera, let us review some fundamentals about the industrial vision system.

Fundamentals of vision system

All AOI original equipment manufacturers (OEMs) provide specifications about their products. Let us consider an example of such specification related to vision performance (Fig. 4).

It is very important to understand the meaning of each parameter to visualise the capability of AOI.

AOI process starts with a digital image. Capturing an image or picture represents a scene. Here picture/scene means it is a combination of object + shape + background (Fig. 5). The quality of the picture depends on brightness, sharpness, and clarity.

- Pixel = Picture + element
- Resolution = Number of pixels/inch
- Pixel dimension = 16 × 12 pixels
- Pixel density = 16 pixels/inch = Resolution

In digital language, a picture is a matrix of pixels. A pixel can be defined as the smallest element in a picture, which means the smallest unit of the entire image. Each pixel location rep-